ANANNYA POPAT

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EDUCATION

University of Toronto — **Sep 2023 - Jan 2025**
Master's, Applied Computing, AI Concentration — *GPA: 3.94*

PROFESSIONAL EXPERIENCE

NextPathway — **Toronto, ON, Canada**
AI Engineer (Automation Lead) — *Aug 2025 - Present*
- Analyzed client requirements and migration code across diverse ETL and data platforms, including Informatica, SQL, Spark, and DataStage.
- Designed and implemented LLM-powered multi-agent pipelines using LangGraph and OpenAI APIs to crawl and translate legacy codebase to Snowflake/GCP-powered systems with 100% accuracy - boosting migration performance by 40% and reducing manual effort by 50%.
- Developed a synthetic data generation engine to create test data for evaluating the performance of translated/migrated code-bases.
- Enabled the organization to attain Snowflake's highest partner designation and drive multi-million-dollar revenue growth across enterprise client engagements.

University Health Network (UHN) — **Toronto, ON, Canada**
Machine Learning Specialist (Team Lead) — *Feb 2025 - Aug 2025*
- Engineered an interactive web-based 3D anatomical model from CT scans using MedSAM2, TotalSegmentator, VTK and GenAI to enhance surgical planning and improve patient outcomes. Utilized on-prem NVIDIA GPUs to fine-tune models and run experiments.
- Integrated fine-tuned GramGAN model into backend via Docker containers to apply realistic medical textures onto anatomical structures.
- Implemented a Tool-Tissue Interaction (TTI) detection model using YOLOv11 (transfer learning) and fine-tuned Vision Transformer (ViT) model for precise tool and interaction classification in the OR to advance surgical practice in real-time.
- Fine-tuned ResNet50 model to classify histologic types in early gastric cancer (EGC) from endoscopic images, achieving 91% specificity for undifferentiated types and 87% specificity for differentiated type

University Health Network (UHN) — **Toronto, ON, Canada**
AI Research Intern — *May 2024 - Dec 2024*
- Led the development of an interactive and realistic 3D anatomical model from patient-specific CT scans to enhance surgical planning.
- Leveraged nnU-Net based TotalSegmentator for precise tissue segmentation and implemented 3D modeling using Visualization Toolkit (VTK) for interactive rendering.
- Performed a comparative study on texture extraction and mapping for medical images using Wavelet Transform, Gabor Filters, fine-tuned Neural 3D Style Transfer, and fine-tuned GramGAN, where deep learning techniques performed 61.4% better than traditional methods.

AdGlobal360 — **Gurugram, HR, India**
Data Science Intern — *May 2022 - Jul 2022*
- Developed an lead scoring prediction model using XGBoost to identify potential buyers based on website activity with 95% F1 score.
- Worked with big data and performed exploratory data curation and analysis using SQL, Spark and Python frameworks for accessing client data entries for ML modeling.

KEY PROJECTS

(AnvayaAI) Explanimator: Research paper to Animated Videos - ***Link to project*** — **Toronto, ON, Canada**
Founder and Lead Engineer (Personal Project)
- Designed an agentic AI system that transforms research papers into intuitive visual explanation videos using manim engine
- Implemented multi-agent architecture for explanation generation, LLM (Gemini 3.1 Pro) judge and manim code generation.
- Paired explanation generation agent with LLM-judge to evaluate content on 6 heuristics, designed after iterative prompt engineering.
- Paired code generation agent with execution-based Python and Manim renderer to ensure the code is runnable.
- Implemented failure-recovery module for code generator by pairing with RAG over Manim documentation using ChromaDB for grounding.
- Integrated beat-level OpenAI TTS narration and dynamically synchronized animation timing with audio, adjusting render speeds to ensure precise audiovisual alignment.

Optimized LLM and RAG Modeling: Classification and Instruction Fine-Tuning - ***Link to project*** — **Toronto, ON, Canada**
Personal Project
- Integrated attention mechanisms like causal, multi-query, and grouped multi-query attention with KV cache optimization.
- Experimented with positional embedding strategies in Transformers such as sinusoidal, learned, and relative position embeddings.
- Implemented classification and instruction fine-tuning of GPT2 using different prompt styles and data processing techniques.
- Deployed on AWS Sagemaker using an MLOps pipeline for automated training, validation, and deployment.
- Built a RAG agent (Agentic AI) for Pokemon Battle Analysis using Gemini API, LangChain and Streamlit

Pocket Comedian: BosonAI Hackathon Winner - ***Link to project*** — **Toronto, ON, Canada**
BosonAI Hackathon Project

- Developed a real-time “pocket comedian” application using Audio LLMs, Flask and Socket.IO, enabling seamless speech-to-speech comedic interactions in an engaging UI.
- Integrated the Higgs Audio Understanding Model for speech input processing, Qwen3 for generating context-aware comedic responses, and the Higgs Audio TTS Model for expressive comedic response.
- Implemented features such as response-voice selection based on your favorite comedian, different comedic modes, multi-turn conversation memory for better banter and integration of Mandarin and French input and audio responses.

Ink-To-Tint: Manga Artisan - *Link to project* **Toronto, ON, Canada**
Master's Academic Project, University of Toronto
- Automated manga colorization and style conversion to enhance readability and ease artists’ workload.
- Optimized image processing techniques like dodging and dilation to decolorize colored manga datasets.
- Trained a Pix2Pix conditional GAN from scratch with quantization to colorize manga pages with a 55% decrease in MSE loss.
- Applied knowledge distillation using pre-trained Stable Diffusion model (MeinaMix v10) to style transfer across four distinct manga styles.

Text-based 3D Gaussian Splatting Object Segmentation - *Link to project* **Toronto, ON, Canada**
Master's Academic Project, University of Toronto
- Developed a 3D Gaussian Splatting segmentation model using LangSAM for multi-model text-driven 3D segmentation.
- Devised an optimized prompt initialization strategy employing K-means clustering for optimal view selection and point sampling.
- Improved IoU by 3%, accuracy by 1%, and reduced computational load by 50% while maintaining ear-optimal results.
- Improved results for the paper ”SAGD: Boundary-Enhanced Segment Anything in 3D Gaussian via Gaussian Decomposition".

InstantSplat Implementation: Unbounded Sparse-view Pose-free Gaussian Splatting - *Link to project* **Toronto, ON, Canada**
University of Toronto
- Implemented key concepts from the InstantSplat paper preceding official code release to enable optimized 3D Gaussian Splatting.
- Replaced traditional SfM (COLMAP) pipelines with a transformer-based pose estimation method inspired by DUSt3R.
- Reconstructed 3D scenes efficiently using joint optimization of camera extrinsics and gaussians and fewer input images under 40 seconds!

Qualitative Badminton Player Detection and Analysis - *Link to project* **Vellore, TN, India**
Bachelor's Capstone Project, Vellore Institute of Technology
- Detected the exact boundary points of the badminton court using Probabilistic HoughTransform and K-means Clustering.
- Tracked near and far-side badminton players in videos using the Particle Filter algorithm with 99% accuracy.
- Predicted badminton strokes (net-drop return, defense, smash) of near-side player using Pillow and CNNs, with 81% accuracy, with the aid of pose estimation algorithm using OpenPos.

SKILLS

Skills: Python, R, Java, SQL, HTML/CSS, Pytorch, LangChain, LangGraph, Tensorflow, OpenCV, Scikit-learn, Streamlit, Flask, Blender, Git, Linux/Unix, Docker, Computer Vision, 3D Computer Vision, Large Language Models, Deep Learning, Machine Learning, Data Engineering, Natural Language Processing (NLP), Google Cloud Platform, AWS Sagemaker, Microsoft Azure, Snowflake, Apache Kafka, Kubernetes, NumPy, Pandas, C/C++, Blender, VTK, NiBabel, Spark, REST APIs, NLTK, Hadoop, React.js, Node.js

PUBLICATIONS AND CONFERENCES

Movie Poster Genre Classification using Federated Learning, ICMLDE 2022, Elsevier
Published paper (11 citations): doi.org/10.1016/j.procs.2023.01.177
- Designed a decentralized federated architecture for movie genre classification, achieving 81% accuracy with local CNN training, enhanced privacy and reduced storage requirements (Google Scholar Profile).

An optimized handwritten polynomial equations solver using an enhanced inception V4 model, *Multi-media Tools and Applications 2023, Springer*
Published Paper (9 citations): doi.org/10.1007/s11042-023-17574-1
- Web-based system that uses an enhanced Inception V4 CNN to recognize and solve handwritten polynomial equations. The model is trained on data from MathNet (arithmetic symbols), MNIST (digits), and EMNIST (alphabet characters) (Google Scholar Profile).

Histology Classification for Early Gastric Cancer using AI Model, SAGES 2025 *Conference Presentation, UHN*
- Fine-tuned ResNet50 model to classify histologic types in early gastric cancer (EGC) from endoscopic images, achieving 91% specificity for undifferentiated types and 87% specificity for differentiated type (Google Scholar Profile).